Assignment 3

Digital Image Processing - M25

Deadline: 25th October , 2025 11:59PM

General Instructions

1. Make sure you run your Jupyter notebook before submitting to save all outputs.
2. If python files are used, make sure to save all the plots in a seperate directory for submission.
3. Answer any questions asked in a markdown cell in your notebook.
4. Proofs can be typed or written.
5. Keep pushing your code as you progress, not just at the end. That is the purpose of using git instead of Moodle.

Q1. Image Classification (50 Marks)

Problem Statement

Design and implement a complete image classification pipeline for handwritten digits(60k images from MNIST dataset) **using only digital image processing methods** (no machine learning classifiers or learned feature representations). Your pipeline should take as input a grayscale image of a digit and output a predicted class in $\{0, 1, \ldots, 9\}$. You are free to explore and combine any classical image processing techniques learnt during the course(filtering, morphological operators, frequency-domain features, moments, shape descriptors, etc.). The dataset can be downloaded from this link or the train data directly from `torchvision.datasets.MNIST`
Note : You are allowed to use *cv2* library for individual components of your pipeline. It is necessary to vectorize your code for improving speed.

DO NOT push the dataset to git

Some Hints (optional to use)

- The magnitude spectrum of the FFT is translationally invariant (up to phase) and radial summaries of the magnitude can capture some interesting features which could be used for preprocessing.
- Autocorrelation and its maxima reveal repeating stroke patterns and symmetry.
- Simple shape descriptors (area, perimeter, Euler number, number of holes) are surprisingly discriminative for many digit pairs.

Implementation and Metrics (40 points)

Implement your full pipeline and evaluate on the MNIST test split. For reproducibility, state any preprocessing parameters and the decision rules you used.

You must report:

- Overall classification accuracy on the MNIST test set.
- Per-class accuracy for each digit 0–9 (present this as a table).
- Confusion matrix (visualize as a heatmap and include it in your write-up).

Analysis and Reasoning (10 points)

Provide a detailed analysis of the results, including:

- What image properties or statistics your method exploits (
- A careful inspection of the confusion matrix: which digit pairs are most often confused and why (give visual/structural explanations).
- What worked well and what failed along with reasoning.

Note: It is recommended that you document everything for your own benefit but it suffices to document the plots and metrics and explain the analysis in the eval.

Q2. Maximum Value of Auto-correlation (Mathematical Proof) (10 Marks)

Problem Statement

Let $f(x,y)$ and $g(x,y)$ be two distinct real-valued images defined on the same finite discrete grid. Define the normalized cross-correlation $R_{fg}(u,v)$ and the normalized auto-correlation $R_{ff}(u,v)$ at shift (u,v) by

$$R_{fg}(u,v) = \frac{\sum_{x,y} \left[f(x,y) - \overline{f}_{uv}\right] \left[g(x-u,y-v) - \overline{g}\right]}{\sqrt{\sum_{x,y} \left[f(x,y) - \overline{f}_{uv}\right]^2 \sum_{x,y} \left[g(x-u,y-v) - \overline{g}\right]^2}} \tag{1}$$

Prove that for any two (not necessarily orthogonal) distinct images f and g,

$$\max_{u,v} R_{ff}(u,v) \geq \max_{u,v} R_{fg}(u,v).$$

Hint - Cauchy Schwarz inequality

Note : You are free to use your notation, make sure to define it clearly at the beginning.

Q3. Image Restoration(30 Marks)

Problem statement

Motion blur is a common degradation in images captured with camera motion or moving objects. In this assignment you will design, implement, and evaluate deblurring methods to recover a sharp latent image $f(x, y)$ from an observed blurred and noisy image $g(x, y)$. The forward model is:

$$g(x, y) = (f * h)(x, y) + n(x, y),$$

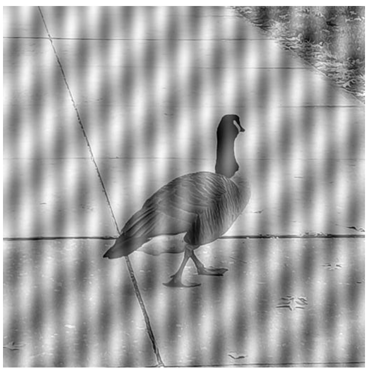
where $h(x, y)$ is the point-spread function (PSF) describing motion blur, $*$ denotes 2D convolution, and $n(x, y)$ is additive noise. For this question, you will restore 2 images.

You may **use any method necessary** for image retrieval. The provided images are:

- A motion-blurred grayscale image $g(x, y)$
- The PSF $h(x, y)$ of the motion blur.
- *Bonus:*(8 Marks) if the PSF is *not* provided, estimate the PSF from $g(x, y)$ and justify your estimation method. you will receive bonus points which are only applicable if you lost marks in other components of the assignment.

Hint : You can explore wiener filter and other frequency domain filtering techniques.
Note : You need not get the perfect restoration, get the possible output that you can.

Q4. Filtering (10 Marks)



Identify and remove the artifacts from the images to reveal the underlying scene. Explain the approach and steps you would take to restore the image and discuss the specific image processing techniques involved in correcting such issues.